

Lab 1 – GradPace Product Description

Stephen Steyaert

Old Dominion University

CS411W

Professor Hosni

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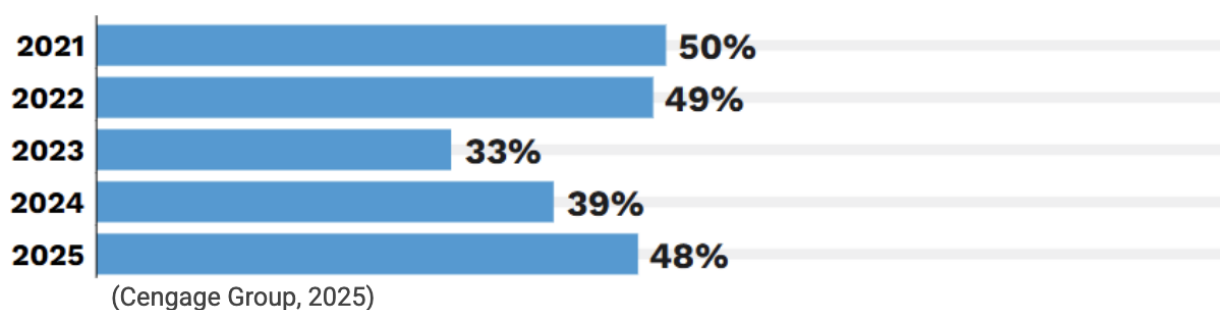
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1 Introduction

Workforce expectations are rapidly evolving, driven by rapid technological advancement and the increasing integration of AI across industries. As a result, employers are facing significant challenges in identifying candidates with the right skills, with 63% citing skill gaps as a top barrier to growth (World Economic Forum, 2025). At the same time, organizations are adjusting their hiring strategies—66% of enterprises report reducing entry-level hiring due to AI, while 91% indicate that automation is actively changing or eliminating jobs (Laurent, 2026). These shifts are contributing to widespread employment challenges for recent graduates, with an estimated 50–60% experiencing underemployment and 48% feeling underqualified for entry-level roles (Cengage Group, 2025). As illustrated in Figure 1, perceptions of preparedness among recent graduates continue to fluctuate, reflecting uncertainty in how well higher education aligns with workforce demands.

Figure 1:

Fluctuations in Perception of Preparedness in Recent Graduates



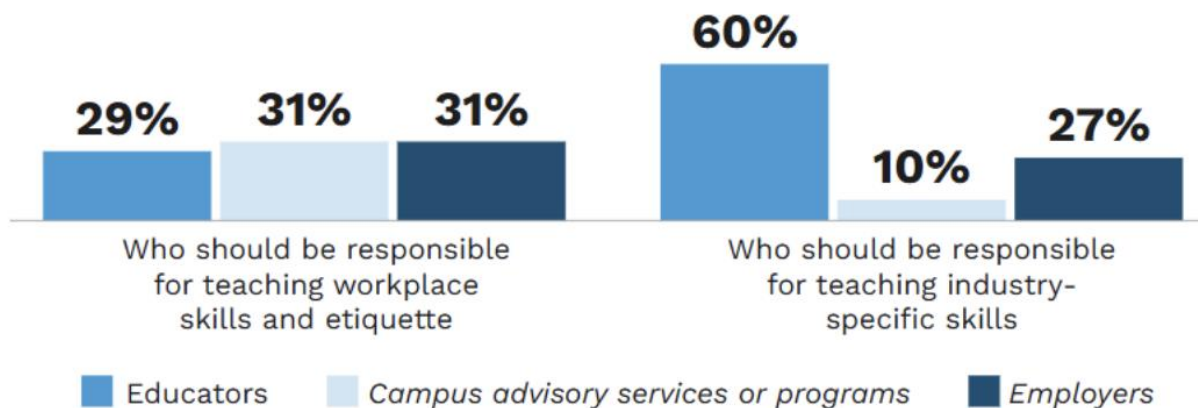
This problem is not isolated to the job market alone but extends across higher education institutions and career services. While academic progress is typically well-tracked, career preparation remains fragmented. Existing tools tend to address isolated needs—such as coursework tracking systems that focus only on academic data, job boards that list opportunities

without personalized guidance, and networking platforms that emphasize connections without structured development. This disjointed ecosystem creates a critical gap: there is no unified, real-time system to guide students in building and assessing their career readiness.

Several key characteristics define this problem. Career development systems rely on fragmented data, making it difficult to form a complete picture of a student's preparedness. Gap detection is often delayed, occurring only after missed opportunities or unsuccessful applications. As a result, career decision-making becomes reactive rather than proactive, and students have limited visibility into their true readiness for the workforce. From an educator's perspective, as shown in Figure 2, there is also ongoing ambiguity around the role institutions should play in career preparation versus academic instruction.

Figure 2:

Educator POV on Teaching Responsibilities



(Cengage Group, 2025)

An effective solution to this problem would need to bring these fragmented sources of information together into a single, continuously updated view of each student's academic and professional progress. It would also need to move career guidance from a passive resource that

students must seek out to an active one that surfaces relevant, time-sensitive opportunities, such as internships, certifications, and networking events; based on each student's own skills and goals. Finally, it would need to benchmark student profiles against real industry and employer expectations so that skill gaps can be identified and quantified early, giving students a clear, evidence-based measure of their readiness rather than a guess.

To address these challenges, GradPace is introduced as a unified, AI-driven career readiness platform designed to bridge the gap between academic progress and professional development, providing students with real-time insights, personalized roadmaps, and actionable guidance toward achieving career success

2 Product Description

GradPace is a web-based career readiness platform that connects a student's academic progress with real-time professional opportunities. Students build a profile from their coursework, grade point average (GPA), internships, certifications, and extracurricular activities, and an AI-driven analysis engine compares that profile against industry and employer benchmarks to identify skill gaps as they emerge, rather than after a student has already missed an opportunity.

The goal of GradPace is to replace the fragmented, reactive approach to career preparation described in the Introduction with a single, proactive system. Its objectives are to give students continuous visibility into their career readiness, to surface personalized and time-sensitive recommendations, such as relevant coursework, certifications, internships, and local networking events, and to give institutions a data-driven way to understand where their career-preparation resources are succeeding or falling short.

2.1 Key Product Features and Capabilities

GradPace provides students with a personalized career roadmap generated from their academic and professional data. Students input their coursework, GPA, internships, certifications, and extracurricular activities, and the system continuously compares this profile against industry and employer benchmarks to produce a readiness score and an interactive roadmap of recommended next steps.

What distinguishes GradPace from existing tools, such as coursework trackers, job boards, and networking platforms, is that it unifies these previously separate functions into one system and evaluates them together using AI. Rather than requiring students to interpret job postings or alumni advice on their own, GradPace's gap-detection engine flags missing skills or credentials as soon as they can be identified, and its recommendation engine translates those gaps into concrete, prioritized actions pulled in part from location-based event data.

By continuously scoring readiness and surfacing opportunities before they are missed, GradPace turns career preparation from a reactive, self-directed search into an ongoing, guided process. This directly addresses the core problem identified in the Introduction: students currently have no unified, real-time way to see how prepared they are for the workforce, and GradPace closes that gap by keeping academic progress and professional opportunity constantly in view side by side.

2.2 Major Components (Hardware/Software)

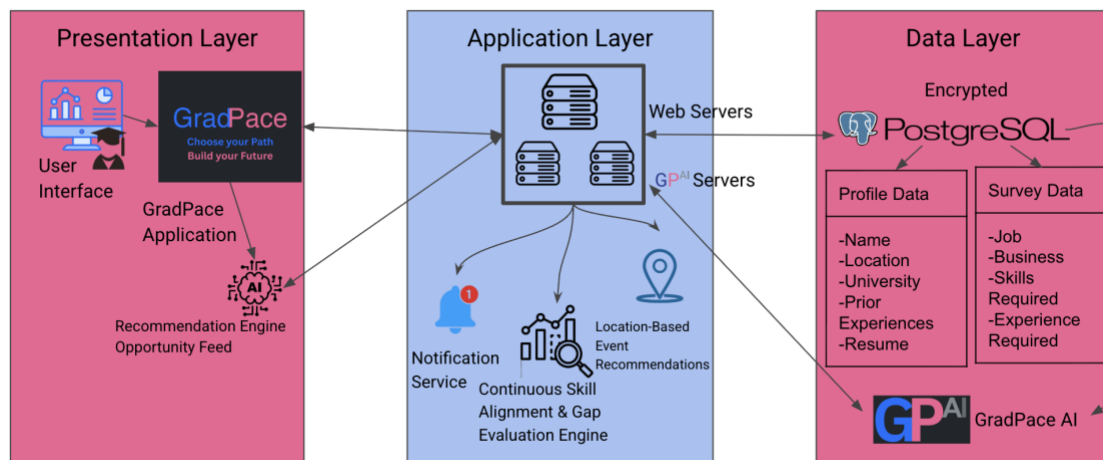
GradPace is deployed as a cloud-hosted web application rather than a locally installed program. Its infrastructure consists of redundant, load-balanced web servers that handle student-

and administrator-facing traffic, and a separate set of AI servers that run the recommendation and gap-detection models. Encrypted student profile data (academic records, prior experience, resumes) and survey and benchmark data (job, business, and skill requirements gathered from alumni and industry input) are stored in a dedicated data layer accessible to both server groups.

As shown in Figure 3, the system is organized into three layers. The presentation layer is the web application itself: the unified academic and opportunity view, the readiness dashboard (progress, gaps, and benchmarks), the opportunity feed and alerts, and the recommendation and relevance engine that generates each student's next steps.

Figure 3:

GradPace Major Functional Components Diagram



The application layer runs on the web server and a dedicated GradPace AI server, and hosts the user/alumni profile manager (registration, login, and preferences), the continuous skill-alignment and gap-evaluation engine (career readiness scoring and benchmark aggregation), the proactive guidance and notification service, and the location-based event recommendation logic that filters and ranks opportunities against each student's gaps. The data layer stores encrypted profile and

survey data in PostgreSQL and connects to integration APIs for external event feeds, with the trained GradPace AI model itself also residing in this layer.

The backend will be built with Django and the Django REST Framework (DRF) in Python, using PostgreSQL for persistent storage and DRF SimpleJWT for authentication. The frontend will be a TypeScript, ReactJS, and Vite application. The AI-driven analysis relies on the OpenAI API for gap detection and recommendation generation, with the Google Maps and Events APIs supplying location-based opportunity data and Postfix handling outbound email notifications. Development will use Git and GitHub Actions for version control and continuous integration, with Pytest and Jest covering the backend and frontend respectively.

3 Identification of Case Study

GradPace is being developed for undergraduate students, initially undergraduate computer science majors, who must independently connect their coursework to evolving market expectations without a unified, data-informed system. As described in the Introduction, these students are most exposed to the problem GradPace addresses: career preparation resources exist, but they are fragmented, passive, and disconnected from a student's actual academic progress, so gaps in skills or experience are often not discovered until it is too late to act on them.

Within that broader population, four groups are most directly affected and inform the case study. Early undergraduates must navigate DegreeWorks, Canvas, and Handshake separately, without a continuous view of their own progress. Upperclassmen often do not realize they have skill gaps until their junior or senior year, because they have lacked proactive guidance up to that point. Ambitious students want more structure around how their current progress compares to industry expectations, and career-focused students struggle with undefined benchmarks, unsure of what being “ready” for their target career actually looks like. The

prototype's case study group will be a small cohort of undergraduate computer science students at Old Dominion University, recruited to build profiles, use the roadmap and gap-detection features, and provide feedback on the accuracy of their recommendations and readiness scores.

To verify these recommendations rather than rely on students' self-reported progress alone, the study will also include two Old Dominion University academic advisors and three to four industry contacts from Hampton Roads technology employers. The academic advisors will review each participant's GradPace-generated roadmap and confirm that the suggested courses and certifications actually exist and align with the university's course offerings, catching any AI-generated recommendation that turns out to be inaccurate or infeasible. The industry contacts will supply the benchmark data the gap-detection engine measures students against and will periodically review a sample of student profiles to confirm the flagged skill gaps and readiness scores reflect what they would actually look for in a candidate. Participating benefits each role directly: students gain concrete, externally verified feedback on where they stand for their target roles; advisors gain visibility into where their advisees are falling behind before it affects their job prospects; and industry contacts gain early insight into a pipeline of students actively closing the skill gaps that matter to their own hiring needs. Together, these three roles let the study exercise GradPace's core capabilities end to end, profile creation, gap detection, roadmap generation, and location-based event recommendations, with the advisor and industry review acting as a check on the accuracy of the AI's benchmarking rather than students grading their own progress. The study will run across a single academic semester, with the goal of showing that participating students close identified skill gaps and report higher readiness confidence by the end of the term than they did at the start.

To see how this works for an individual student, consider a sophomore computer science major in the case study cohort who is targeting a backend developer role. She creates her GradPace profile by entering her current coursework, GPA, and a summer internship, then selects backend development as her target career path. GradPace's gap-detection engine compares her profile against the employer benchmarks collected for that role and flags that she has no coursework or experience in database design, a requirement most local employers list for the position. Her readiness dashboard updates to show this gap alongside her overall readiness score, and the recommendation engine responds by surfacing a database elective offered the following semester along with a local database workshop pulled from GradPace's location-based event data. She adds the elective to her plan and registers for the workshop directly from her opportunity feed. Over the following weeks, as she completes the workshop and begins the coursework, she updates her profile, her readiness score rises, and the database gap that once would have gone unnoticed becomes a documented, closed item on her roadmap rather than something she discovers for the first time in a job interview.

Beyond this initial group, GradPace is designed to support additional user roles as the product matures. Alumni can complete surveys that keep industry benchmarks current and receive a portion of their subscription cost back for doing so. Industry professionals can contribute up-to-date qualification and hiring data, which benefits them by surfacing better-prepared candidates. System administrators use the platform to manage accounts, monitor data quality, and maintain the benchmarks the AI relies on. Longer term, GradPace could extend to undergraduates in other majors and to students at other universities facing the same disconnect between academic progress and career readiness.

4 Glossary

API (Application Programming Interface): A set of rules that allows different software applications to communicate and exchange data with each other.

Authentication: The process of verifying a user's identity before allowing access to a system.

Backend: The server-side part of a website or application that handles logic, databases, and data processing.

Career Readiness: A measure of how prepared a student is to enter a specific career based on skills, experience, and qualifications.

Career Roadmap: A structured, visual plan that outlines a student's progress, goals, and required steps toward a specific career path.

Cloud Computing: The use of remote servers on the internet to store data and run applications instead of a local computer.

Data Leak: The accidental or unauthorized exposure of sensitive or private information.

Database: An organized collection of data that can be stored, accessed, and managed electronically.

Encryption: A security method that converts data into a coded form so that only authorized parties can read it.

Frontend: The user-facing part of a website or application that users interact with in their browser.

Gap Detection (Early-Warning System): A feature that identifies missing skills, experiences, or qualifications needed for a target career.

Industry Readiness Benchmarking: The process of comparing a student's profile against expected skills and qualifications for specific careers.

JWT (JSON Web Token): A compact, digitally signed token format used to securely transmit user identity and authorization data between a client and server.

Location-Based Event Recommendations: Suggestions for nearby internships, networking events, or career opportunities based on user location.

Professional Data Input: User-provided information such as coursework, projects, internships, and certifications used to build a profile.

REST (Representational State Transfer): An architectural style for designing web APIs that uses standard HTTP methods to access and manipulate resources.

Survey Results Dashboard: A feature that displays aggregated insights from alumni data to inform students and administrators.

System Administrator: A role responsible for managing the platform, maintaining data quality, and overseeing system functionality.

Token (AI): A small unit of text processed by an AI model, such as a word, part of a word, or punctuation.

User Interface (UI): The visual elements of a software application that allow users to interact with it.

Web Server: A computer system or software that stores websites and delivers web pages to users when they request them through a browser.

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